

JACK DOUGLASS

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SUMMARY

Computational and molecular biologist with startup and operations experience focused on pandemic prevention and reducing existential risk. Currently interning at Hadrian Biodefense, researching how to bolster societal preparedness for infrastructural collapse. Previously investigated and amended virulence factors for high-risk bacteria with the Sequence Biosecurity Risk Consortium, developed DNA filtration software for SecureBio Detection, researched molecular diagnostics for infectious diseases at a Gates Foundation-backed startup, and founded an iGEM team researching organoid development.

EDUCATION

University of Arizona Aug 2023 – Dec 2026
B.S. in Biology | Minors in Computer Science, Biochemistry GPA: 3.96

EXPERIENCE

Hadrian Biodefense (fka ProEquip) Jun 2026 - Present
Research Intern Berkeley, CA

- Investigating the resilience of societal infrastructure during disasters to prepare preparedness recommendations

BlueDot Impact Jun 2026
Operations Contractor London, UK

- Planned and ran on-site operations and logistics for BlueDot Incubator Week v4
- Produced vlog capturing the week's motivation and participant experience

ERA Cambridge Jan 2026 – Mar 2026
AlxBio Fellow Cambridge, UK

- Built a computational pipeline for the **Sequence Biosecurity Risk Consortium (SBRC)** that maps sequence-level variation to biological function and potential hazard, using structural and sequence-based methods including **ESM-C embeddings**
- Applied structured labeling criteria to **NIST and SBRC nucleic acid screening benchmark sets**, comparing screening frameworks and identifying gaps in sequence-of-concern detection methods

Strive Bioscience Dec 2025 – Jun 2026
Research Assistant Oro Valley, AZ

- Sole bioinformatician at a Gates Foundation-backed diagnostics startup; built pipelines for **molecular target selection, probe design, and experimental analysis** for infectious diseases diagnostic development
- Designed and tested nucleic acid diagnostics for tuberculosis, optimizing for **sensitivity, specificity, and stability**

SecureBio Detection Oct 2025 – Jan 2026
Bioinformatics Engineer (Contract) Remote

- Developed open-source bioinformatics software using k-mer filtering, benchmarking **~4x faster** than industry-standard BBDuk and now in production for wastewater-based pandemic surveillance; won **3rd place among university student research** at the 2025 AZBio Awards
- Designed around an **end-to-end biosurveillance pipeline**: how host DNA contamination degrades pathogen signal and how decontamination quality affects downstream detection sensitivity

Arizona iGEM Aug 2025 – Jan 2026
Co-Founder Tucson, AZ

- Founded university iGEM synthetic biology team; recruited a 10-person team, secured lab space and raised \$12K in funding
- Established project direction for iGEM's 2026 competition, focused on **adaptable organoid vascularization systems**

VOLUNTEERING

- Sequence Biosecurity Risk Consortium** – Added and revised sequence of concern definitions for 41 virulence factors
- Effective Altruism Cambridge** – Graded biosecurity projects from the Cambridge High Impact Research Program
- Apart Research** – Judged technical biosecurity projects developed at the 2026 Apart Research AlxBio Hackathon
- Tucson Effective Altruism** – Facilitated UA accredited discussions on catastrophic risks and emerging technology
- Access to Medicines Initiative** – Digitized Nigeria reproductive health clinic data to improve medicine supply chains

SKILLS

Computational: Rust, Python, Java, R, JavaScript, SQL, HTML/CSS, Bash, Git, Docker

Biological: Assay Design and Optimization, Molecular Target Selection and Validation, Bioinformatics

Certifications: Emergency Medical Technician (NREMT), CPR, Laboratory Safety **Languages:** English, Portuguese